BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, MARDAN

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2021

CHEMISTRY

Past Paper 2021 • Paper Code: MRD-X-21 (A)

Retyped faithfully from the original board paper — same questions, no changes.

Section-A

Time Allowed: 20 Minutes | **Maximum Marks:** 12 | **Paper Code:** MRD-X-21 (A) VERSION: C

Note: Note: There are three sections in this paper i.e. A, B & C. Instructions: Please attempt Section-A on the MCQs Answer Sheet only. Only black ball point or marker may be used for shading the relevant circle. Please make sure that there is no cutting, over writing, erasing, or multiple circles shading.

Q1. Choose the correct option i.e. A,B,C, or D.

Q1. The formula of hexane in the shape of $CH_3(CH_2)_4CH_3$ is .....

(A) Condensed
(B) Dot and cross
(C) Structural
(D) Molecular

Q2. Substitution reaction is the characteristic of .....

(A) Alkynes
(B) Alkenes
(C) Alkanes
(D) Acetylene

Q3. Sugars are polyhydroxy derivative of .....

(A) Alkyl halides
(B) Aldehyde
(C) Ketone
(D) Aldehyde & ketone

Q4. 99 % of the atmosphere is made of .....

(A) Nitrogen and oxygen
(B) Argon and carbon dioxide
(C) Hydrogen and neon
(D) Helium and neon

Q5. The ratio between hydrogen and oxygen in water is .....

(A) 2 : 1
(B) 2 : 8
(C) 1 : 8
(D) 2 : 16

Q6. The density of water is maximum at .....

(A) −40°C (B) 40°C

(C) −4°C (D) 4°C

Q7. Forth flotation process is concentration of ore by .....

(A) Magnetically (B) Wetting the ore

(C) Difference in densities (D) Chemically

Q8. Neutral solution has a pH value of .....

(A) 3 (B) 5

(C) 7 (D) 14

Q9. In equilibrium constant Kc = ...

(A) Kf/Kr (B) Ks/Kr

(C) Kr × Kf (D) Kr + Kr

Q10. The bases which are soluble in water are called .....

(A) Acids (B) Alkalis

(C) Salts (D) Amphoteric substances

Q11. Peptide linkage is present in .....

(A) Carbohydrates (B) Lipids

(C) Proteins (D) Vitamins

Q12. The functional group of ketone is .....

(A) −C− with O double-bonded on top (B) −O− with O double-bonded on top

(C) −C−H with O double-bonded on top (D) −C−OH with O double-bonded on top

Section-B

Time Allowed: 2:40 Hours | **Maximum Marks:** 32

Q2. Write short answers to any eight (8) of the following parts. All parts carry equal marks.

(i) Explain Chemical Equilibrium with two examples.

(ii) Compare the physical properties of Acids and Bases.

(iii) Differentiate between Ali-Cyclic and Aromatic organic compound.

(iv) Draw the Molecular, Dot and Cross, Condensed and Structural formula of Ethene (C_2H_4).

(v) Give any four uses of Carbohydrates.

(vi) Differentiate between simple and complex Lipids.

(vii) Give the formation of acid rain from: (a) Carbon Dioxide (CO_2) (b) Sulphur Trioxide (SO_3).

(viii) How Ozone Layer is depleted in the atmosphere? Give the depletion due to oxygen.

(ix) Why water is universal solvent?

(x) Make the distinction between Soft and Hard Water.

(xi) What is Slaked Lime? How it is produced?

Section-C

Maximum Marks: 21

> Note: Answer any three (3) questions. All questions carry equal marks.

Q3.

(a) [4] Discuss Arrhenius concept of Acids and Bases along with its two limitations.

(b) [3] Calculate the pOH and pH of 0.01 M solution of Potassium Hydroxide (KOH).

Q4.

(a) [4] Define Atmosphere and give its composition.

(b) [3] How permanent hardness of water can be removed by Ion Exchange Method?

Q5.

(a) [4] Define refining of Petroleum and give the composition of Petroleum.

(b) [3] Give any three physical properties of Monosaccharides.

Q6.

(a) [4] Explain the graphical presentation of Dynamic Equilibrium along with its examples.

(b) [3] Define functional group and give the functional group of: (i) Alcohol (ii) Aldehyde